generates the gravitational field that governs the shrinkage of rulers and the dilation of the ticking of clocks." Although the two versions of the Einstein field equation are mathematically equivalent, their verbal pictures differ profoundly.

It is extremely useful, in relativity research, to have both paradigms at one's fingertips. Some problems are solved most easily and quickly using the curved spacetime paradigm; others, using flat spacetime. Black-hole problems (for example, the discovery that a black hole has no hair) are most amenable to curved spacetime techniques; gravitational-wave problems (for example, computing the waves produced when two neutron stars orbit each other) are most amenable to flat spacetime techniques. Theoretical physicists, as they mature, gradually build up insight into which paradigm will be best for which situation, and they learn to flip their minds back and forth from one paradigm to the other, as needed. They may regard spacetime as curved on Sunday, when thinking about black holes, and as flat on Monday, when thinking about gravitational waves. This mind-flip is similar to that which one experiences when looking at a drawing by M. C. Escher, for example, Figure 11.2.

Since the laws that underlie the two paradigms are mathematically equivalent, we can be sure that when the same physical situation is analyzed using both paradigms, the predictions for the results of experiments will be identically the same. We thus are free to use the paradigm that best suits us in any given situation.

This freedom carries power. That is why physicists were not content with Einstein's curved spacetime paradigm, and have developed the flat spacetime paradigm as a supplement to it.

Newton's description of gravity is yet another paradigm. It regards space and time as absolute, and gravity as a force that acts instantaneously between two bodies ("action at a distance," Chapters 1 and 2).

The Newtonian paradigm for gravity, of course, is *not* equivalent to Einstein's curved spacetime paradigm; the two give different predictions for the outcomes of experiments. Thomas Kuhn uses the phrase *scientific revolution* to describe the intellectual struggle by which Einstein invented his paradigm and convinced his colleagues that it gives a more nearly correct description of gravity than the Newtonian paradigm (Chapter 2). Physicists' invention of the flat spacetime paradigm was *not* a scientific